

Activity – Week 2

Model Provider	Model	Release Date	Knowledge Cutoff	Token Cost (API)	Context Window	Type	When to use this model
OpenAI	GPT-5.2	Aug-26	Aug-25	See OpenAI API pricing (varies by tier)	~1M tokens	Multimodal, reasoning	General-purpose AI, coding, agents, enterprise assistants
Anthropic	Claude Fable 5	Jun-26	Jan-26	Published API pricing	1M tokens	Multimodal	Long-document analysis, coding, research assistants
Google DeepMind	Gemini 2.5 Flash-Lite	Jun 17, 2025	Early 2025	Low-cost API tier	1M tokens	Multimodal	High-volume inference, chatbots, document processing
xAI	Grok 4	Jul-25	2025	Published via xAI API	256K tokens	Multimodal	Coding, reasoning, real-time knowledge
Mistral AI	Magistral	Jun-25	2025	Published API pricing	128K tokens	Reasoning LLM	Scientific reasoning, enterprise copilots
DeepSeek	DeepSeek-R1	Jan 20, 2025	Mid-2024	Very low-cost API	128K tokens	Reasoning	Math, coding, reasoning at low cost
Alibaba	Qwen3.8-Max	Aug 3, 2026	2026	API pricing announced through Alibaba Cloud	1M tokens	Open-weight multimodal	Large-scale enterprise AI, multilingual applications
Moonshot AI	Kimi K3	Jul 17, 2026	2025	Published API pricing	1M tokens	Open-weight reasoning	Long-context coding and research tasks
Cohere	Command A	Mar-25	2024	Cohere API pricing	256K tokens	Enterprise LLM	RAG, business applications, enterprise search
MiniMax	H3	Jul 31, 2026	2026	Lower-cost commercial pricing	Multimodal long context	Video generation + multimodal	Video generation, advertising, content creation

Transformer

Transformer layers builds on the representations produced by the previous layer, allowing the model to capture increasingly complex relationships in the input before producing the next token

Example for transformer layers

Imagine writing a report that goes through several reviewers:

- **Layer 1:** Fixes grammar.
- **Layer 2:** Improves sentence structure.
- **Layer 3:** Checks facts.
- **Layer 4:** Improves clarity.
- **Layer 5:** Polishes the style.

Self Attention Block - Find which words are important for understanding each word and provides the weightage to the words based on the relationship

Feed Forward Neural Network – Based on the Weightage it will understand the meaning